

Yi ZHU

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EDUCATION

Institut National des Sciences Appliquées de Lyon (INSA Lyon)

Lyon, France

PhD student in Signal and Image Processing

Oct 2024 - Sep 2027

Thesis subject: Creation of digital twins by fusion of radiographic and optical data for oral medicine

Institut National des Sciences Appliquées de Lyon (INSA Lyon)

Lyon, France

Diplôme d'Ingénieur (Equivalent to a Master's Degree in Eng.) in Electrical Engineering

Sep 2019 - Aug 2024

Pohang University of Science and Technology (POSTECH)

Pohang, South Korea

Exchange Student in Electrical Engineering with BRMIE Scholarship

Sep 2022 - Jun 2023

RESEARCH EXPERIENCE

Fusion of IOS and CBCT

CREATIS Laboratory, INSA Lyon

Oct 2024 - Sep 2027

Supervisor: Sébastien Valette, Razmig Kéchichian, Raphaël Richert

- Developing multimodal learning methods for high-fidelity 3D dental reconstruction by fusing Cone Beam Computed Tomography (CBCT) and Intraoral Scan (IOS) data.
- Designed a fusion framework that leverages the complementary strengths of CBCT and IOS: CBCT provides root and internal anatomical information, while IOS provides high-resolution crown surface geometry.
- Investigating deep implicit representations and representation learning strategies to reconstruct patient-specific dental digital twins from incomplete and heterogeneous multimodal data.
- Exploring weakly supervised, semi-supervised, and data-efficient learning methods to reduce the dependence on densely annotated dental datasets.
- Evaluating the proposed methods on public dental datasets and hospital-acquired CBCT/IOS data, with the goal of supporting clinical applications in orthodontics, implantology, and endodontics.

3D Urban Road Reconstruction Using 360° panoramic videos

National Institute of Informatics, Japan

Aug 2025 - Jan 2026

Supervisor: Satoshi Ikehata, Kiyoharu Aizawa

- Implemented and evaluated SOTA algorithms, including VGGT, MapAnything, and Pi3, to assess their applicability in complex urban scene modeling.
- Analyzed the trade-offs between reconstruction accuracy and computational efficiency, identifying critical optimization paths for deploying these models in real-world large-scale environments.

Evaluation of Deep Learning Algorithms for Medical Image Segmentation

CREATIS Laboratory, INSA Lyon

Mar 2024 - Aug 2024

Supervisor: Chantal Muller, Thomas Grenier

- Evaluated segmentation models (YOLOv8/v9, UNeXt, DETR/UNETR, U-Net/nnU-Net) using MONAI's Metrics Reloaded (Dice, Hausdorff, ASSD), with focus on lesion-size performance.
- Pretrained on public datasets, then fine-tuned on custom MS data for improved lesion segmentation.

PUBLICATIONS

Accepted as Poster Presentation, ISBI 2026: "High-Fidelity 3D Tooth Reconstruction by Fusing Intraoral Scans and CBCT Data via a Deep Implicit Representation."

Accepted as Oral Presentation, ICSP 2024: "Comparative Analysis of Three Advanced Deep Learning Algorithms for Multiple Sclerosis Lesion Segmentation in FLAIR MRI."

Submitted to BMVC 2026: "LatentFlowNet: Trajectory-Anchored Latent Dynamics for Target PET Imaging Prediction from Irregular Longitudinal Observations."

Manuscript in preparation for ACCV 2026: "Large-Scale 3D Urban Road Reconstruction."

SKILLS

- Deep Learning & 3D Vision: PyTorch, VGGT, Pi3, U-Net, Point Transformer, DeepSDF
- Tools: Linux, Git, Docker, CUDA, LaTeX
- Languages: English (IELTS 6.5, Linguaskill C1), French (C1), Chinese (Native)